

SEIF OTEFA

647-904-5288 | seifotefa@gmail.com | linkedin.com/in/seif-otefa | github.com/seifotefa | seifotefa.com

EDUCATION

McMaster University

Bachelor's of Applied Science, Computer Science (Honours Co-op)

September 2024 – April 2028 (Expected)

GPA: 3.6/4.0

WORK EXPERIENCE

Founding Software Engineer (Co-op) — Backboard.io

May 2026 – Present

- Led a team of 3 building **Backboard Studio**, an agentic IDE forked from VS Code end to end, architecting three modes: Editor, Simple Mode (chat-first UI for non-technical users), and Advanced, a kanban of concurrent Backboard/Claude/Codex agent sessions.
- Shipped **100+ commits** across dev and prod builds, including a **Rust** bridge to Backboard's API, a marketplace of **34,000+ skills** and **100+ MCP servers**, signed macOS/Windows installers, and in-app updates.
- Built the Backboard Ambassador portal end-to-end, a **React SPA with Python/FastAPI** backend deployed on its own subdomain and **AWS EC2**, with cross-service **SSO**, admin review queues, and private **AWS S3** uploads.
- Launched and ran the **Campus Ambassador program** — owned marketing and content strategy driving **100,000+ impressions**, and interviewed and onboarded the entire founding cohort (**15+ ambassadors**).

Robotics Instructor — O'Botz Robotics

July 2025 – Present

- Taught robotics, electronics, and programming to **10 students** of various ages and skill levels through hands-on robot assembly, coding exercises, and troubleshooting activities.

Software Engineer Intern — Ludera

October 2025 – January 2026

- Built an AI flashcard pipeline using **spaCy**, **PyMuPDF**, and Python to extract PDF/Markdown notes, preprocess raw text, and convert diverse formats into clean, structured data for downstream generation.
- Implemented semantic search with **pgvector** embeddings and evaluated open-source **LLMs (7B–13B)** from Hugging Face for Q&A generation, processing **100+ chunks/hour** with accurate context retrieval and high-quality outputs.

Teaching Assistant — Stanford University

April 2025 – May 2025

- Led weekly **CS106A** sections for **15+ students**, teaching **Python** fundamentals, control flow, graphics, and data structures while providing individualized debugging support and mentorship.

Product Manager Intern — MoVA Realities

March 2025 – May 2025

- Led a cross-functional team of 4 to define an MVP framework for an AI-powered immersive platform, producing a **30-page** requirements analysis outlining **21 core features** and aligning stakeholders through technical documentation, UI/UX mockups, and presentations.
- Developed a **27-week phased roadmap** reducing projected development time by **20%** and designed a comprehensive system architecture mapping **15+ components**, including AI services and cloud infrastructure.

PROJECTS

Frontline – DeltaHacks12 Winner (Best Use of Presage Prize)

React, C++, Tailwind CSS, Node.js, Express.js, Google Gemini API, ElevenLabs, PreSage SDK

- Built a 10-second triage dashboard using Presage (camera-based vitals) and Gemini Vision (injury detection) to generate real-time clinical insights and first aid guidance. **Reached out to by Presage CEO to continue collaboration.**
- Integrated ElevenLabs voice + conversational AI “911 operator” to answer user questions in real time and automatically generate EMS-ready SBAR handoff reports for paramedics.

Notipply – Automated Job Scraping & Instant SMS Notification Platform

React, Tailwind CSS, Node.js, Express.js, Go, Supabase, Clerk API, Stripe API, Twilio API

- Built a real-time job alert platform that scrapes and filters job postings using concurrent **Go** services, delivering instant SMS notifications via **Twilio** based on user-defined preferences.
- Architected backend infrastructure with **Supabase (PostgreSQL)**, implemented authentication with **Clerk**, and integrated Stripe for subscription monetization, validating demand with **25+ users** on the pre-launch waitlist.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Rust, Go, Java, C, C++, SQL, HTML/CSS, Bash, PHP

Frameworks & Tools: React, Node.js, FastAPI, Electron, Tailwind, spaCy, Git/GitHub, Claude Code, Codex

Cloud, Databases & AI: AWS (EC2, S3), MySQL, PostgreSQL, Redis, LLM APIs (OpenAI, Gemini, Claude), RAG, MCP, REST APIs, CI/CD, Agile